Machine Learning Classification of Inflammatory Bowel Disease from Gut Microbial Taxonomic Abundance

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***Abstract*—Inflammatory bowel disease (IBD), including Crohn's disease (CD) and ulcerative colitis (UC), is a chronic gastrointestinal condition associated with substantial clinical, social, and economic burden. This project evaluates whether gut microbial taxonomic abundance profiles can be used to classify IBD status using supervised machine learning. The final dataset was drawn from the Integrative Human Microbiome Project 2 / Inflammatory Bowel Disease Multi'omics Database (HMP2 / IBDMDB) and combined public metadata with metagenomic taxonomic profiles. After filtering to metagenomics samples and merging the files by sample identifier, the final modeling table contained 1,638 samples from 130 participants. CD and UC were collapsed into a binary IBD class and compared against nonIBD controls. Preprocessing included participant-aware train/test splitting to prevent longitudinal leakage, genus-level feature extraction, prevalence filtering, and log1p transformation. Multiple classifiers were compared, including Logistic Regression, Random Forest, Extra Trees, Support Vector Machine, Gradient Boosting, and K-Nearest Neighbors. Gradient Boosting achieved the best ROC-AUC and, after threshold tuning, produced a more balanced tradeoff between IBD recall and nonIBD specificity. These results suggest that gut microbiome taxonomic abundance contains useful predictive signal for IBD classification, while also highlighting limitations related to class imbalance, compositional data, repeated longitudinal sampling, and lack of external validation.**

***Index Terms*—inflammatory bowel disease, microbiome, machine learning, classification, metagenomics, taxonomic abundance**

I. Introduction

Inflammatory bowel disease (IBD) refers primarily to Crohn's disease (CD) and ulcerative colitis (UC), both of which involve chronic inflammation of the gastrointestinal tract. IBD is lifelong, recurrent, and often expensive to manage. It affects quality of life, requires repeated medical follow-up, and can lead to hospitalization, surgery, and long-term medication use. In the United States, the disease burden is substantial. The U.S. Centers for Disease Control and Prevention (CDC) estimates IBD prevalence at roughly 2.4 to 3.1 million people and reports approximately $8.5 billion in annual IBD-related health care costs in 2018 [1].

The burden extends beyond direct medical costs. Adults with IBD report higher rates of psychological distress and other chronic comorbidities than adults without IBD [1]. Work productivity impairment is also substantial, with pooled estimates of 16.41% absenteeism, 35.91% presenteeism, and 39.41% overall productivity loss in IBD [2]. These facts make IBD a meaningful machine learning problem rather than an abstract classification exercise.

The goal of this project was to test whether microbial taxonomic abundance from stool metagenomic samples can classify IBD versus nonIBD status using supervised learning. The final task was binary classification, with CD and UC combined into one positive IBD class and nonIBD retained as the negative class. Because false negatives are more costly than false positives in a screening-oriented context, recall was treated as a high-priority metric, while specificity and balanced accuracy were also emphasized to avoid misleading conclusions from class imbalance.

II. Related Work

Prior microbiome research strongly supports the biological relevance of this project. Lloyd-Price et al. used the HMP2 / IBDMDB cohort to show that IBD is associated with broad microbial, biochemical, and host-level dysbiosis, establishing HMP2 as a major public resource for IBD microbiome analysis [3]. Morgan et al. reported that IBD-related microbial dysfunction involves both taxonomic and functional changes, reinforcing the idea that microbiome profiles contain disease-relevant signal [4]. Gevers et al. further showed that treatment-naive Crohn's disease is associated with clear microbial shifts, strengthening the biological plausibility of microbiome-based classification [5].

Machine learning precedent also informed the project design. Vangay et al. introduced the Microbiome Learning Repo (MLRepo), which framed microbiome prediction as a reproducible machine learning task family and motivated the Milestone 1 starting point of the project [6]. Manandhar et al. applied supervised learning to fecal microbiome data for IBD diagnosis and showed that models such as Random Forest can achieve useful discrimination, supporting the use of multiple supervised classifiers and metrics such as ROC-AUC, precision, recall, and confusion matrices [7]. Nguyen et al. systematically reviewed IBD machine learning studies and noted that many models suffer from limited validation and risk of bias [8]. That finding directly influenced the present project's emphasis on leakage control and careful interpretation.

Microbiome abundance data also create methodological challenges. Gloor et al. argued that microbiome sequencing data are compositional rather than absolute, meaning that relative abundance values must be interpreted cautiously [9]. Quinn et al. similarly emphasized compositional reasoning in omics analysis [10]. These studies informed two decisions in the present work: first, to avoid overstating biological conclusions from feature importance, and second, to use simple, defensible preprocessing rather than making strong claims about causal microbial mechanisms.

III. Dataset

The final dataset came from the Integrative Human Microbiome Project 2 / Inflammatory Bowel Disease Multi'omics Database (HMP2 / IBDMDB). Two public files from the same study resource were used: `hmp2_metadata_2018-08-20.csv` and `taxonomic_profiles_3.tsv.gz`. The metadata file contains participant- and sample-level descriptors, including diagnosis and participant identifiers, while the taxonomic profile file contains metagenomic microbial taxonomic abundance data. The public metadata release used in this project is dated 2018-08-20, and the underlying cohort is longitudinal, meaning that some participants contributed multiple samples over time [3].

Before task-specific filtering, the metadata file contained 5,533 rows and 490 columns. After restricting the analysis to metagenomics samples and removing rows missing required identifiers or diagnosis labels, the labeled metagenomics metadata contained 1,638 samples. The taxonomic profile file contained 932 taxa/features across 1,639 columns, from which 187 genus-level features were retained before prevalence filtering. After merging metadata and taxonomy by sample identifier, the final modeling table contained 1,638 metagenomic samples from 130 unique participants, including 750 CD samples, 459 UC samples, and 429 nonIBD samples. For the final prediction task, CD and UC were collapsed into a binary IBD class, resulting in 1,209 IBD samples and 429 nonIBD samples.

Table I summarizes the size, class structure, and modeling dimensions of the final HMP2 / IBDMDB dataset after preprocessing.

Figure 3 shows the binary target distribution together with a PCA projection of the genus-level abundance matrix. The bar chart shows that IBD is the majority class, while the PCA projection shows broad overlap between groups, indicating that the classification problem is not trivially separable by eye.

This dataset has clear strengths and limitations. Its major strength is scale relative to the Milestone 1 benchmark-style dataset, which was much smaller. It also contains participant identifiers, which allow a leakage-safe participant-aware split. However, the dataset is longitudinal, class-imbalanced toward

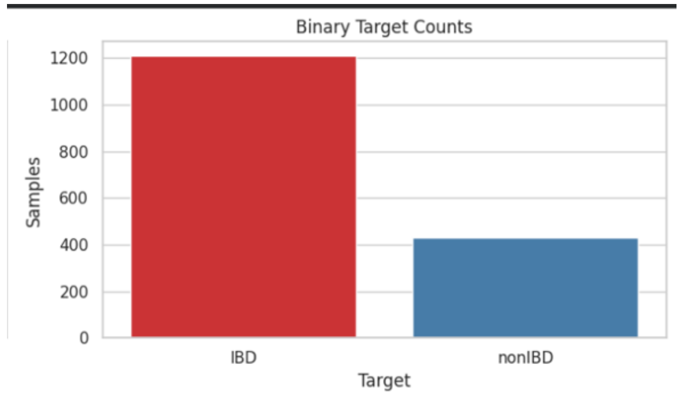

Fig. 1. Bar chart consisting of CD, UC being absorbed to a binary classification of IBD vs nonIBD.

TABLE I
Dataset Summary

Item	Value
Full metadata shape	5,533 rows × 490 columns
Labeled metagenomics metadata	1,638 samples
Taxonomic profile table	932 taxa/features × 1,639 columns
Merged modeling table	1,638 samples × 678 columns
Unique participants	130
Diagnosis counts	750 CD, 459 UC, 429 nonIBD
Binary target counts	1,209 IBD, 429 nonIBD
Raw genus features	187
Selected genus features	78
Train samples	1,333
Test samples	305
Train participants	104
Test participants	26
Participant overlap	0

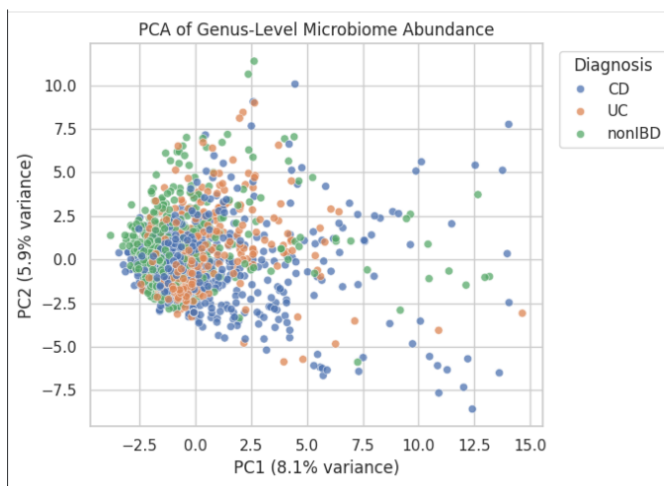

Fig. 2. Binary target distribution and PCA of genus-level microbiome abundance. The class distribution confirms that IBD is the majority class, while the PCA projection shows overlap between groups despite visible feature structure.

IBD, and still limited to a single cohort. These features shaped the preprocessing and evaluation strategy.

IV. Methods

A. Milestone 1 Background and Dataset Change

Milestone 1 originally used a smaller microbiome benchmark-style dataset derived from the MLRepo / Morgan 2012 direction. The recovered local Milestone 1 working subset contained 74 merged samples and 3,677 microbial features, including 59 Crohn's disease samples and 15 healthy samples. That milestone established the problem framing and early exploratory analysis, but the dataset was too small for strong final claims. As a result, the project pivoted to HMP2 / IBDMDB for the final submission.

B. Preprocessing

The final preprocessing pipeline was designed to be reproducible and leakage-safe. The major steps were:

1) Load the HMP2 metadata and keep metagenomics samples only.
2) Map the diagnosis labels into a binary target: nonIBD to 0, CD to 1, and UC to 1.
3) Load the taxonomic abundance file and retain genus-level taxa only.
4) Convert taxonomic paths into readable genus labels.
5) Match taxonomy sample identifiers to metadata sample identifiers.
6) Convert feature values to numeric and fill invalid or missing abundance values with zero.
7) Split the data by participant rather than by sample row.
8) Fit a prevalence filter on the training data only, retaining features present in at least 5% of training samples.
9) Apply log1p transformation to reduce skew and sparsity effects.

The average zero fraction across the selected microbial features was approximately 63.43%, indicating a sparse abundance matrix. This is typical for microbiome data and justified both feature filtering and transformation.

C. Participant-Aware Splitting

Because HMP2 is longitudinal, naive row-random splitting would risk placing samples from the same participant into both training and test sets. In that scenario, a model could partially memorize participant-specific microbial patterns and appear to generalize better than it truly does. To prevent this, `GroupShuffleSplit` was used with participant identifier as the grouping variable. This ensured zero participant overlap between train and test sets and made the final evaluation more defensible.

D. Models and Evaluation

The following models were compared on the held-out test set:

- Majority baseline
- Logistic Regression
- Random Forest
- Extra Trees
- Support Vector Machine with RBF kernel
- Gradient Boosting
- K-Nearest Neighbors

The majority baseline was included specifically to show why raw accuracy can be misleading in an imbalanced dataset. Because IBD is the majority class, a trivial classifier that predicts everything as IBD can appear strong by accuracy while completely failing to identify nonIBD controls.

The project reported accuracy, balanced accuracy, precision, recall, specificity, F1 score, ROC-AUC, precision-recall AUC, and confusion matrix counts. Recall was emphasized because false negatives are especially costly in this setting. Missing a true IBD case is more harmful than flagging a nonIBD case for additional follow-up. At the same time, specificity and balanced accuracy were also important because a model that predicts nearly everything as IBD is not clinically useful.

After the best non-baseline model was identified by ROC-AUC, the classification threshold was tuned on a validation split derived from the training data only. Thresholds from 0.05 to 0.95 were scanned, and the final threshold was selected among thresholds with recall at least 0.60, favoring higher balanced accuracy and then higher F1 score.

V. Results

Exploratory analysis confirmed that the merged dataset was imbalanced toward IBD. The class distribution plot showed that IBD was the majority class, and the PCA projection showed broad overlap between groups, indicating that the classification problem was not trivially separable by eye but still contained enough structure to justify supervised learning.

Table II compares the baseline and supervised classifiers on the held-out test set. The majority baseline achieved an accuracy of 0.839 and recall of 1.000, but its specificity was 0.000. This was a useful warning: a model can appear strong by raw accuracy simply by predicting nearly everything as IBD.

TABLE II
Comparative Model Performance on the Test Set

Model	Acc.	Bal. Acc.	Prec.	Recall	Spec.	ROC-AUC
Gradient Boosting	0.800	0.600	0.871	0.895	0.306	0.763
Extra Trees	0.797	0.615	0.876	0.883	0.347	0.681
Random Forest	0.784	0.533	0.849	0.902	0.163	0.706
SVM (RBF)	0.728	0.640	0.891	0.770	0.510	0.689
Logistic Regression	0.603	0.582	0.877	0.613	0.551	0.656
Majority Baseline	0.839	0.500	0.839	1.000	0.000	0.500
KNN	0.636	0.453	0.822	0.723	0.184	0.475

Among the non-baseline models, Gradient Boosting achieved the best ROC-AUC and strong recall. However, the default threshold still produced poor specificity. Threshold tuning identified a cutoff of 0.67 and improved the tradeoff between IBD detection and nonIBD recognition.

Table III shows the final tuned Gradient Boosting metrics on the untouched test set.

Figure 4 shows the ROC curve for the final Gradient Boosting model. The ROC-AUC of 0.763 indicates meaningful

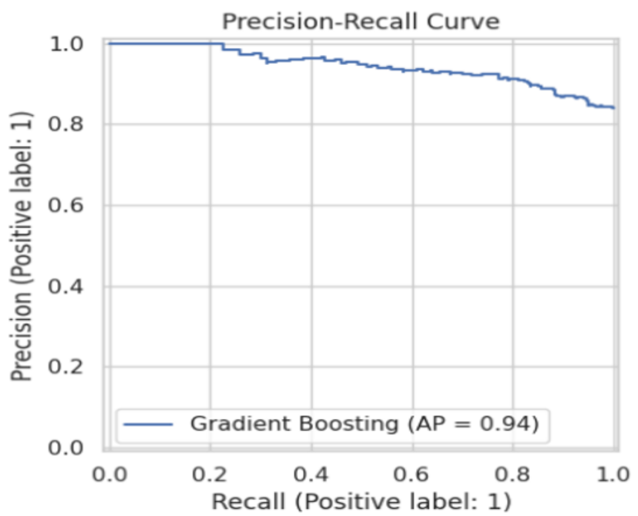

Fig. 3. Especially useful in imbalanced classification because it focuses on positive-class detection quality .

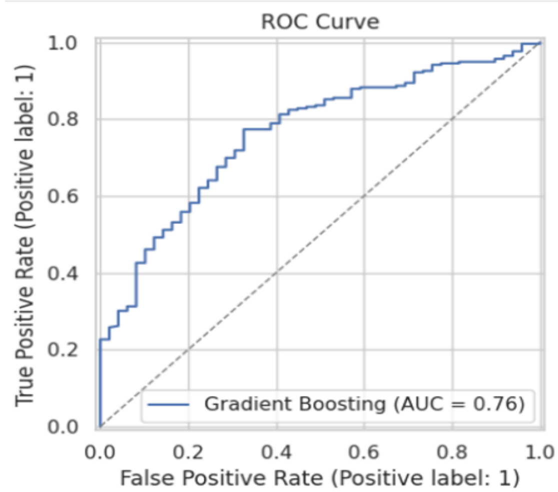

Fig. 4. ROC curve for the final Gradient Boosting model. The curve summarizes the tradeoff between true positive rate and false positive rate across thresholds and shows discrimination above random guessing.

TABLE III
FINAL TUNED GRADIENT BOOSTING PERFORMANCE

Metric	**Value**
Accuracy	0.764
Balanced accuracy	0.694
Precision	0.911
Recall	0.797
Specificity	0.592
F1 score	0.850
ROC-AUC	0.763
Precision-recall AUC	0.944
True negatives	29
False positives	20
False negatives	52
True positives	204

discrimination above random guessing and supports the choice of Gradient Boosting as the final model.

In addition to ROC-AUC, the final tuned model achieved precision 0.911, recall 0.797, specificity 0.592, and balanced accuracy 0.694. These values show that the final model still detected most IBD cases while improving healthy-control recognition relative to the default-threshold behavior.

VI. Discussion and Limitations

The final results suggest that microbial taxonomic abundance contains useful predictive signal for IBD classification. In plain language, the best final model could correctly identify many IBD cases while also improving its ability to avoid falsely labeling nonIBD controls as positive once the classification threshold was tuned. This does not mean the model is ready for clinical use, but it does support the broader idea that stool microbiome profiles are informative enough to justify further machine learning investigation.

Several limitations remain. First, the analysis used one final dataset and did not include external validation on an independent cohort, so generalizability remains uncertain. Second, the feature space is sparse and compositional, which complicates interpretation and may affect model behavior. Third, the longitudinal structure of the dataset required careful splitting and still creates methodological complexity. Fourth, the binary target was imbalanced toward IBD, making evaluation sensitive to threshold choice and metric selection. Finally, the feature importance outputs are predictive rather than causal and should not be overinterpreted biologically.

VII. Future Work

Future work could strengthen this project in several directions. The most important next step would be external validation on an independent IBD microbiome cohort. Other improvements could include comparing binary classification with a multiclass CD versus UC versus nonIBD task, incorporating additional omics layers from HMP2, using more formal calibration and hyperparameter search procedures, and exploring participant-level longitudinal models rather than treating each sample primarily as a static example. Additional work could also compare simpler transformations against more explicitly compositional-data-aware methods.

VIII. Conclusion

This project demonstrated that supervised machine learning can use gut microbial taxonomic abundance profiles to classify IBD versus nonIBD in a large public microbiome cohort. The final HMP2 / IBDMDB dataset provided a much stronger basis for analysis than the smaller Milestone 1 benchmark direction. Gradient Boosting achieved the best ranking performance, and threshold tuning improved the tradeoff between IBD recall and nonIBD specificity. Overall, the project supports the conclusion that microbiome-based IBD classification is a meaningful research problem, while also reinforcing the need for careful preprocessing, participant-aware validation, and cautious interpretation.

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